→ TCL (Transaction Control Language).
→ DCL (Data Control Language).

DDL

This consists of the SQL commands that can be used to define the database structure or schema.

List of DDL commands

→ CREATE : Creates the new database or its objects
→ ALTER : Alters the structure of the database.
→ DROP : Deletes objects from the database.
→ TRUNCATE : Removes all records from a table.
→ RENAME : renames a object in the database.

SYNTAX → CREATE TABLE <table name>
unique name
(<column1> <data type(length)> [column constraint],
<column 2> <data-type (length)> [column constraint],
... <column n> <data-type (length)> [column constraint]

* We are creating a table with a unique table name & in the bracket specifies the column names along with their data types. [column constraints] is optional.

Ex:- CREATE TABLE record (
student_id SERIAL PRIMARY KEY,
name VARCHAR (100),
age INT,
dept VARCHAR (50)
);
ALTER TABLE record ADD COLUMN
specialization VARCHAR(100);
Select * from record.

* The above code creates columns with the given titles in the CREATE TABLE & adds a column specialization to it.

* To add a primary key
ALTER TABLE <table name> ADD PRIMARY KEY (id);

* RENAME COLUMN : Used to change the name of a column.

Ex:- ALTER TABLE <table name>
RENAME COLUMN OOPS TO JAVA;

*DROP : Deletes all the database objects.

Ex:- DROP TABLE <table name>.

* TRUNCATE: Used to delete all the data from the table

Syntax → TRUNCATE TABLE table-name;

*ALTER TABLE RENAME command : Changes the name of existing table.

Ex:- ALTER TABLE person RENAME TO employee;

DML (Data Manipulation Language)

→ SELECT ⇒ Retrieve data from tables.
→ INSERT ⇒ Add new rows into a table.
→ UPDATE ⇒ Modify existing records in a table
→ DELETE ⇒ Remove rows from a table.

*Used to access, store, modify, update & delete.

DCL (Data Control Language)

1.) Grant DCL Command
*Used to give user access privileges to a database.

2.) Revoke DCL command
* This is used to remove all the permissions applied by the Grant command.

Transaction Control Language (TCL)

* These commands can only be used with DML commands.

→ Commit ⇒ Allows the database users to save the operations in the database.

→ Roll back ⇒ Undo's transaction that have not already been saved to the database.

PostgreSQL Data Types

① Numeric

→ smallint, integer, bigint, decimal, serial.

② Character

→ char, varchar, character

③ Date/Time Datatype.

→ date, interval, timestamp.

④ Monetary

→ money.

⑤ Binary

→ bytea

⑥ Boolean

→ boolean

⑧ Geometric

→ point, line, circle.

⑨ Text Search

→ tsvector

→ tsquery

⑩ Network Address

→ inet

→ cidr.

→ macaddr.

Constraints in PostgreSQL

→ Not Null ⇒ ensures a column cannot have a null value.

→ Check

→ Unique ⇒ ensures all values in a column of a table are exclusive.

→ Primary Key ⇒ specifies each row or record in a database table uniquely & make sure that there is no record duplicity.

→ Foreign Key ⇒ used to define that the value in a column or field of a table is equal to primary key of another table.

→ Exclusion ⇒ ensures that any two rows are linked on the specified columns or statements using the defined operators. In any case, one of these operators will return null or false.

Types of SQL Functions

→ Group Functions

* Functions act on set of values are known as group functions.

Ex:- SUM, AVG, MIN, MAX

→ Scalar Functions

* Functions act on only one value at a time known as scalar fn.

Ex:- Length, ASCII.

AVG Function

* Returns Average of all row values for particular column.

Syntax → AVG([DISTINCT | ALL] column_name);

MIN

* Returns a minimum value of Expressi

Syntax → MIN([DISTINCT | ALL] column_name);

MAX

* Returns maximum value present in particular column of Data table.

Syntax → MAX([DISTINCT | ALL] column_name).

COUNT

* Returns number of rows present in a column.

Syntax → COUNT( );

SUM

* Returns sum of values in column.

* Syntax → SUM();

ABS Function

ABS(n) ⇒ returns absolute value.

Ex:- Select ABS(-10)

O/p:- 10

POWER

POWER (m,n);

Ex:- Select POWER(6,2);

O/P:- $6^2 = 36$

ROUND

ROUND (m,n) → numeric value with decimal point; n → rounded position.

Ex:- Select ROUND(10.29,1)

O/P:- 10.3

SQRT

SQRT(n)

Ex:- SQRT (25)

= 5

EXPONENT

* Returns e raised to nth Power.

→ EXP(n)

→ e^n

GREATEST

GREATEST (21,10,30)

O/P:- 30

LEAST

LEAST (35,75,25)

O/P:- 25

MOD

MOD(m,n) Gives remainder Ex:- MOD(16,7)

TRUNC

TRUNC (number, decimal places);

Ex:- select TRUNC(17.235, 1)

O/P:- 17.2

FLOOR

FLOOR(n)

Ex:- FLOOR(24.18)

O/P:- 24

CEIL

CEIL(n)

Ex:- CEIL(24.84)

O/P:- 25

STRING FUNCTIONS

LOWER

LOWER(Char);

Ex:- LOWER('XYZ')

O/P:- xyz

UPPER

UPPER(Char);

Ex:- UPPER('xyz')

O/P:- XYZ

INITCAP

INITCAP(Char);

Ex:- INITCAP('abc')

O/P:- Abc

SUBSTRING

SUBSTRING(String, start-position, length);

Ex:- SUBSTRING('SEKHAR', 1

O/P:- SEKHA

ASCII

→ ASCII (char);

Ex:- ASCII ('a')

O/P:- 97.

POSITION

POSITION (Search string in main string).

POSITION ('EKH' in 'SEKHAR')

O/P:- 2.

LENGTH

LENGTH (word);

Ex:- LENGTH ('SEKHAR');

O/P:- 6

Ltrim

LTRIM (char, set);

ltrim('xyz', 'x')

O/P:- yz.

Rtrim

RTRIM (char, set);

Ex:- rtrim ('xyz', 'z')

O/P:- xy

LPAD

LPAD (char1, n, [char2]);

Ex:- lpad('xyz', 5, *)

O/P:- **xyz.

RPAD

RPAD (char1, n, [char2]);

Ex:- rpad ('xyz', 6, 'p')

O/P:- xyzppp.

CONCAT

CONCAT (argument1, argument2);

Ex:- CONCAT ('abc', 'XYZ')

O/P:- abcXYZ.

NESTED SUB QUERY

→ A subquery is a query within another SQL query & embedded within the WHERE clause.

→ = > < >= <= <> != <=>

* Can appear in WHERE, FROM, SELECT

Types Based on Execution

→ Single-row subquery ⇒ Only one value.
(=, >, <)
→ Multi-row subquery ⇒ returns multiple values.
(IN, ALL, ANY)
→ Correlated subquery ⇒ inner query depends on outer query row.

JOINING Relations

* JOIN means to combine two or more tables.

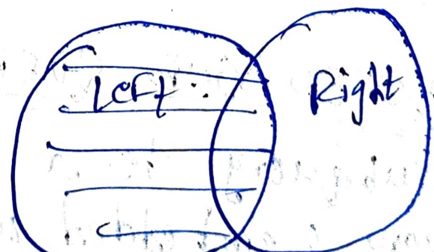
1) INNER JOIN

* This keyword selects records that have matching values in both tables.

Syntax → SELECT column_name
FROM table 1
INNER JOIN table 2
ON table1.column_name = table2.column_name;

2) Left Outer Join

* Returns all records from the left table, and the matching records from the right table.

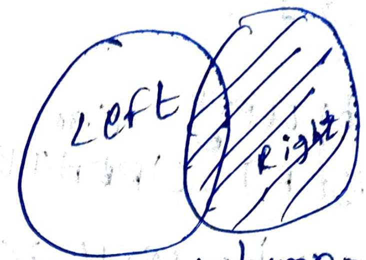


Syntax → SELECT column_name
FROM table 1
LEFT JOIN table 2
ON table1.column_name = table2.column_name.

Right Outer Join

Returns all records from right table & matching records from the left table.

Syntax→ SELECT column-name
FROM table1
RIGHT JOIN table2
ON table1.column-name = table2.column-name;



Full Outer Join

returns all records when there is a match in left or right table records.

Syntax→ SELECT column-name
FROM table 1
FULL OUTER JOIN table 2
ON table1.column-name = table2.column-name
WHERE condition;

Cross Join

No of rows multiplied in the fist table by a number of rows in second table.

Syntax→ SELECT column-name
FROM table 1
CROSS JOIN table 2;

Note:- CREATE FUNCTION statement is used to develop user-defined functions.

* "inc" function

SELECT inc(20);

O/P:- 21

SELECT (inc(inc(20))

O/P:- 22

* Subquery can be nested inside a SELECT, INSERT, UPDATE, DELETE.
* A subquery is usually added within the WHERE clause of another SQL SELECT statement.

syntax → Select select_list
From table
Where expr operator

* The subquery executes once before the main query executes.
* The main query use the subquery result

Relational Query Languages

* Allow manipulation & retrieval of data from a database.

BASIC OPERATIONS

Select : σ
Project : π
Union : $\cup$
Set difference : $-$
Cartesian product : $\times$
rename : ρ
Joins